

P2P Multicast System - Complete Project Document

A full reference document containing the problem statement, system design, architecture, all source code, documentation, and test suite for the Decentralized Peer-to-Peer Multicast System.

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1. Problem Statement

Background

In traditional networked systems, nodes rely on a centralized server or tracker to discover peers, register themselves, and maintain a list of active participants. This introduces a single point of failure - if the central server goes offline, the entire network loses coordination capability.

Problem

Design and implement a fully decentralized peer-to-peer (P2P) network where:

- Nodes can discover each other automatically without any central server or registry.
- Each node independently maintains a list of active peers and registered data/channels.
- The network is resilient - stale or disconnected nodes are automatically detected and removed.
- Node state is persisted so a peer can rejoin the network and recover its previous configuration.

Constraints

- Use only Python standard library (socket, threading, json, hashlib, os, time).
- Communication must happen via UDP Multicast to support broadcast-style local network discovery.
- The system must handle concurrent access safely with threading locks.
- Each node's configuration must be saved and restored across restarts.

Solution

A P2P Multicast System where every node:

- Broadcasts its presence via UDP Multicast (group 224.1.1.1:5007) every 5 seconds.
- Listens for other nodes' broadcasts and registers them as peers.
- Runs a health monitor that removes peers silent for more than 10 seconds.
- Saves and loads its complete state from a local JSON config file.
- Provides a CLI shell for manual node registration and peer inspection.

2. System Overview

Property	Value
Language	Python 3.8+
Protocol	UDP Multicast
Multicast Group	224.1.1.1
Multicast Port	5007
Buffer Size	4096 bytes
Discovery Interval	Every 5 seconds
Peer Timeout	10 seconds
Hashing Algorithm	SHA-256 (via hashlib)
Config Storage	storage/configs/{port}.json
Dependencies	None (stdlib only); pytest for testing

Key Features

- ? Fully decentralized - no central server
- ? Auto-discovery via UDP multicast broadcast
- ? Persistent state - configs survive restarts
- ? Health monitoring - dead peers auto-pruned
- ? Thread-safe - all shared state protected by locks
- ? Modular architecture - clean separation of concerns
- ? Tested - 7 unit tests covering all core components

3. Directory Structure

```
p2p_multicast_system/  
?  
??? main.py           # Application entry point  
?  
??? core/             # Core logic package  
?   ??? __init__.py  
?   ??? node.py        # P2PNode orchestrator class  
?   ??? discovery.py   # UDP multicast sender & listener threads  
?   ??? health_monitor.py # Peer heartbeat monitoring & pruning  
?   ??? config_manager.py # JSON config load/save manager
```

```

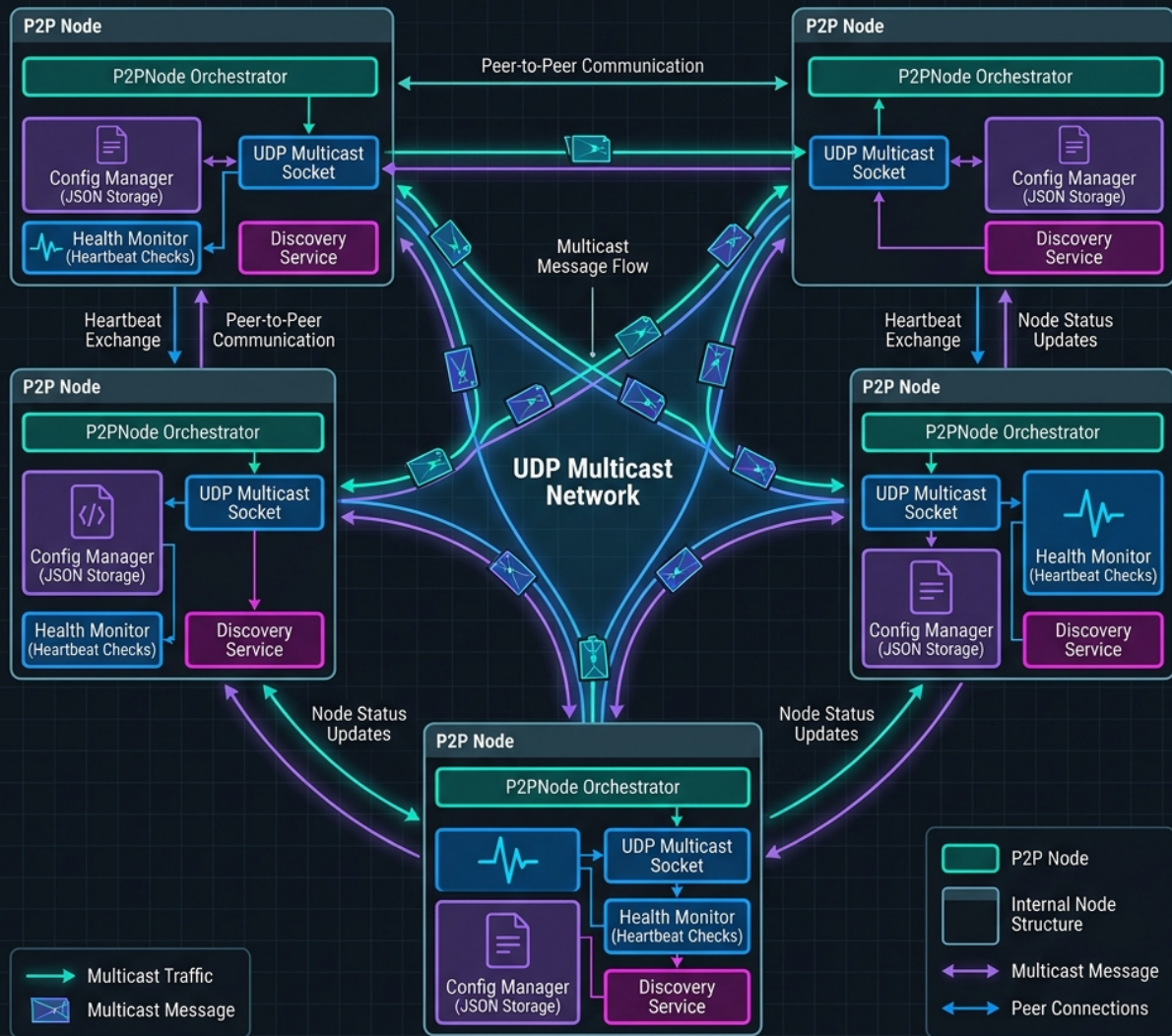
?
??? utils/                                # Shared utilities package
?   ??? __init__.py
?   ??? hashing.py                        # SHA-256 hash generation utility
?   ??? constants.py                      # Network settings and timeout values
?
??? storage/
?   ??? configs/                          # Dynamic JSON peer state files
?       ??? 5001.json                     # Auto-created when node on port 5001 runs
?       ??? 5002.json                     # Auto-created when node on port 5002 runs
?       ??? ...
?
??? docs/
?   ??? design_document.md                # Architecture and design decisions
?   ??? architecture_diagram.png          # Visual system diagram
?   ??? workflow.md                       # Step-by-step interaction sequences
?
??? tests/
?   ??? test_hashing.py                   # Unit tests for hashing utility
?   ??? test_discovery.py                 # Unit tests for discovery threads
?   ??? test_health_monitor.py            # Unit tests for health monitor logic
?
??? requirements.txt                      # Testing dependencies
??? README.md                            # Project setup and usage guide

```

4. Architecture Diagram

DECENTRALIZED P2P MULTICAST NETWORK ARCHITECTURE

MULTICAST DATA FLOW & COMPONENT INTERACTION



5. Design Document

Architecture Overview

The system is a decentralized peer-to-peer (P2P) network where nodes discover and register each other dynamically using UDP Multicast. No central tracker or coordinator is required.

The system is split into modular components:

- **P2PNode** (`core/node.py`) - The central coordinator. Sets up UDP sockets, handles user CLI, manages shared state, and initializes sub-services.
- **ConfigManager** (`core/config_manager.py`) - Encapsulates all disk I/O. Automatically stores node state in `storage/configs/{port}.json` relative to the project root.
- **Discovery Service** (`core/discovery.py`):
 - **Sender Loop** - Periodically broadcasts node presence via UDP multicast packets (every 5 seconds).
 - **Listener Loop** - Continuously listens for other nodes' multicast broadcasts and registers them.
- **Health Monitor** (`core/health_monitor.py`) - Runs every 2 seconds. Removes any peer whose `last_seen` timestamp is older than `PEER_TIMEOUT` (10 seconds).
- **Utilities** (`utils/`) - Independent, reusable utility functions and global constants.

Network Design

Property	Detail
Multicast IP	224.1.1.1 - Administratively scoped local multicast add
Transport Port	5007
Protocol	UDP - lightweight, connectionless, suitable for broadca
Message Format	JSON-encoded byte payloads
Socket Option	SO_REUSEADDR - allows multiple sockets to bind to the s
Multicast Join	IP_ADD_MEMBERSHIP - joins the multicast group on all in

DISCOVER Message Format

```
{
  "type": "DISCOVER",
  "peer_id": "192.168.1.10:5001",
  "host": "192.168.1.10",
  "port": 5001,
  "nodes": ["channel_a", "channel_b"],
  "hash": "e3b0c44298fc1c149afb..."
}
```

Config File Format (storage/configs/{port}.json)

```
{
  "peer_id": "192.168.1.10:5001",
  "host": "192.168.1.10",
  "port": 5001,
  "registered_nodes": ["channel_a"],
  "peers": {
    "192.168.1.10:5001": {
      "host": "192.168.1.10",
      "port": 5001,
      "nodes": ["channel_a"],
      "hash": "e3b0c44298fc1c149afb...",
      "last_seen": 1716812345.678
    },
    "192.168.1.11:5002": {
      "host": "192.168.1.11",
      "port": 5002,
      "nodes": ["channel_b"],
      "hash": "a87ff679a2f3e71d9181...",
      "last_seen": 1716812348.123
    }
  }
}
```

Design Decisions

Decision	Rationale
UDP over TCP	Discovery is stateless broadcast; TCP overhead is unnec
SHA-256 peer hash	Provides a stable, verifiable fingerprint for peer stat
threading.Lock	Ensures safe concurrent access to peers and registered_
list(peers.items()) in monitor	Prevents RuntimeError from dict size changes during ite

Config saved to storage/configs/	Keeps state files organized and out of the root directory
Daemon threads	Ensures background threads don't block process exit

6. Workflow

Node Initialization Sequence

When a node starts:

- Port Prompt - User enters a valid port number ($0 < \text{port} < 65535$).
- Identity Creation - Resolves machine IP -> constructs `peer_id = "IP:Port"`.
- Config Loading - Checks for `storage/configs/{port}.json`:
 - Found -> Restores `registered_nodes` and `peers`.
 - Not found -> Creates a fresh config file.
- Socket Setup - Creates UDP socket, sets `SO_REUSEADDR`, binds to multicast port, joins multicast group.
- Self Registration - Adds itself to `peers` dict with current timestamp.
- Thread Launch - Starts 3 daemon threads: listener, sender, health monitor.
- CLI Shell - Enters the interactive command loop.

Peer Discovery Sequence

```
sequenceDiagram
    autonumber
    participant NodeA as Node A (Port 5001)
    participant MC as Multicast Group 224.1.1.1:5007
    participant NodeB as Node B (Port 5002)

    Note over NodeA,NodeB: Both nodes running their send/listen loops

    NodeA->>MC: Broadcasts DISCOVER {peer_id, host, port, nodes, hash}
    MC->>NodeB: Delivers DISCOVER message

    Note over NodeB: peer_id != self.peer_id -> accepted
    Note over NodeB: Updates peers["192.168.x.x:5001"] with last_seen=now()
    NodeB->>NodeB: Saves state to storage/configs/5002.json

    NodeB->>MC: Broadcasts DISCOVER {peer_id, host, port, nodes, hash}
    MC->>NodeA: Delivers DISCOVER message

    Note over NodeA: peer_id != self.peer_id -> accepted
    Note over NodeA: Updates peers["192.168.x.x:5002"] with last_seen=now()
    NodeA->>NodeA: Saves state to storage/configs/5001.json
```

Health Monitor Pruning Sequence

```
Every 2 seconds:
????????????????????????????????????????
? Acquire threading.Lock                ?
? current_time = time.time()            ?
? For each peer_id in peers:            ?
?   if peer_id == self.peer_id: skip    ?
?   if (current_time - last_seen)       ?
?       > PEER_TIMEOUT (10s):          ?
?       -> mark for removal              ?
? Delete all marked peers                ?
```

```
? If any removed: save_config() ?
? Release Lock ?
????????????????????????????????????????
```

CLI Commands

Command	Action
register	Prompts for a channel/node name string and associates i
view nodes	Lists all registered nodes across all discovered peers
view peers	Shows Peer ID, IP, port, and first 20 chars of SHA-256
exit	Gracefully shuts down the node

7. Source Code

main.py

```
import sys
import os

# Add the directory containing main.py to sys.path to allow execution
# without needing extra environment variables (PYTHONPATH) setup.
sys.path.append(os.path.dirname(os.path.abspath(__file__)))

from core.node import P2PNode

def main():
    """Main entrypoint for starting a P2P Multicast node."""
    try:
        node = P2PNode()
        node.start()
    except KeyboardInterrupt:
        print("\nShutdown signal received. Exiting peer node...")
        sys.exit(0)

if __name__ == "__main__":
    main()
```

core/node.py

```
import socket
import threading
import time
from utils.constants import MULTICAST_GROUP, MULTICAST_PORT
from utils.hashing import generate_hash
from core.config_manager import ConfigManager
from core.discovery import start_discovery_sender, start_discovery_listener
from core.health_monitor import start_health_monitor

class P2PNode:
    """
    Orchestrator class for the Peer-to-Peer Multicast node.
    Manages socket initialization, command-line interface, configuration state,
```

```

and runs discovery and health monitoring services.
"""

def __init__(self):
    self.host = socket.gethostbyname(socket.gethostname())
    self.port = self.get_port()
    self.peer_id = f"{self.host}:{self.port}"

    self.config_manager = ConfigManager(self.port)
    self.registered_nodes = []
    self.peers = {}
    self.running = True
    self.lock = threading.Lock()

    self.load_config()

    # Create UDP Multicast socket
    self.sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM, socket.IPPROTO_UDP)
    self.sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)

    try:
        self.sock.bind(("", MULTICAST_PORT))
    except Exception:
        self.sock.bind((MULTICAST_GROUP, MULTICAST_PORT))

    # Join the Multicast Group
    mreq = socket.inet_aton(MULTICAST_GROUP) + socket.inet_aton("0.0.0.0")
    self.sock.setsockopt(
        socket.IPPROTO_IP,
        socket.IP_ADD_MEMBERSHIP,
        mreq
    )

def get_port(self) -> int:
    """Prompts the user for a valid TCP/UDP port number."""
    while True:
        try:
            port = int(input("Enter port: "))
            if 0 < port < 65535:
                return port
            print("Invalid port")
        except Exception:
            print("Enter numeric value")

def save_config(self):
    """Saves node's current status and peers to disk."""
    self.config_manager.save_config(
        self.peer_id,
        self.host,
        self.registered_nodes,
        self.peers
    )

def load_config(self):
    """Loads node configuration from disk if available."""
    config = self.config_manager.load_config()
    if config:
        self.registered_nodes = config.get("registered_nodes", [])
        self.peers = config.get("peers", {})
        print(f"\nLoaded config from {self.config_manager.config_file}")
    else:
        self.save_config()

```



```

def register_node(self):
    """Interactive helper to register a new user-defined node/channel."""
    node_name = input("\nEnter node string: ").strip()
    if node_name == "":
        print("Invalid node")
        return

    with self.lock:
        if node_name not in self.registered_nodes:
            self.registered_nodes.append(node_name)
            self.peers[self.peer_id] = {
                "host": self.host,
                "port": self.port,
                "nodes": self.registered_nodes,
                "hash": generate_hash(self.peer_id),
                "last_seen": time.time()
            }
            self.save_config()
            print(f"\nRegistered node: {node_name}")
        else:
            print("Node already exists")

def view_nodes(self):
    """Prints all registered nodes across all discovered peers."""
    with self.lock:
        print("\n===== ALL REGISTERED NODES =====")
        for peer_id, info in self.peers.items():
            print(f"\nPeer : {peer_id}")
            for node in info["nodes"]:
                print(f"  - {node}")
        print("\n=====")

def view_peers(self):
    """Prints information about all currently active peers."""
    with self.lock:
        print("\n===== PEERS =====")
        for peer_id, info in self.peers.items():
            print(f"""
Peer ID : {peer_id}
IP      : {info['host']}
Port    : {info['port']}
Hash    : {info['hash'][:20]}...
""")
        print("=====")

def command_loop(self):
    """Loop that runs in the main thread accepting user commands."""
    while self.running:
        try:
            command = input(f"\n[{self.port}] Enter command: ")
            if command.lower() == "register":
                self.register_node()
            elif command.lower() == "view nodes":
                self.view_nodes()
            elif command.lower() == "view peers":
                self.view_peers()
            elif command.lower() == "exit":
                self.running = False
                break
        except (KeyboardInterrupt, EOFError):
            self.running = False
            break

```

```

def initialize_self_peer(self):
    """Ensures the node registers itself in its own peer map."""
    with self.lock:
        self.peers[self.peer_id] = {
            "host": self.host,
            "port": self.port,
            "nodes": self.registered_nodes,
            "hash": generate_hash(self.peer_id),
            "last_seen": time.time()
        }
    self.save_config()

def start(self):
    """Starts all daemon service threads and enters command shell loop."""
    self.initialize_self_peer()
    print(f"\nPeer started on {self.peer_id}")

    start_discovery_listener(self)
    start_discovery_sender(self)
    start_health_monitor(self)

    self.command_loop()

```

core/discovery.py

```

import socket
import json
import time
import threading
from utils.constants import MULTICAST_GROUP, MULTICAST_PORT, DISCOVERY_INTERVAL, BUFFER_SIZE
from utils.hashing import generate_hash

def start_discovery_sender(node) -> threading.Thread:
    """
    Starts a background daemon thread that broadcasts the node's info
    periodically via UDP multicast.
    """
    def send_loop():
        while node.running:
            message = {
                "type": "DISCOVER",
                "peer_id": node.peer_id,
                "host": node.host,
                "port": node.port,
                "nodes": node.registered_nodes,
                "hash": generate_hash(node.peer_id)
            }
            try:
                node.sock.sendto(
                    json.dumps(message).encode(),
                    (MULTICAST_GROUP, MULTICAST_PORT)
                )
            except Exception:
                pass
            time.sleep(DISCOVERY_INTERVAL)

    thread = threading.Thread(target=send_loop, daemon=True)
    thread.start()

```

```

        return thread

def start_discovery_listener(node) -> threading.Thread:
    """
    Starts a background daemon thread that listens for discovery messages
    from other peers and registers them.
    """
    def listen_loop():
        while node.running:
            try:
                data, addr = node.sock.recvfrom(BUFFER_SIZE)
                message = json.loads(data.decode())

                # Ignore discovery messages from self
                if message["peer_id"] == node.peer_id:
                    continue

                if message["type"] == "DISCOVER":
                    with node.lock:
                        node.peers[message["peer_id"]] = {
                            "host": message["host"],
                            "port": message["port"],
                            "nodes": message["nodes"],
                            "hash": message["hash"],
                            "last_seen": time.time()
                        }
                    node.save_config()
            except Exception:
                pass

    thread = threading.Thread(target=listen_loop, daemon=True)
    thread.start()
    return thread

```

core/health_monitor.py

```

import time
import threading
from utils.constants import PEER_TIMEOUT

def start_health_monitor(node) -> threading.Thread:
    """
    Starts a background thread that monitors peer activity.
    If a peer has not been seen (no DISCOVER message received) for more than
    PEER_TIMEOUT seconds, it is removed from the active peer list.
    """
    def monitor_loop():
        while node.running:
            current_time = time.time()
            remove_peers = []

            with node.lock:
                # Copy dictionary keys/items to avoid dict size mutation during iteration
                for peer_id, info in list(node.peers.items()):
                    if peer_id == node.peer_id:
                        continue

                    if current_time - info["last_seen"] > PEER_TIMEOUT:
                        remove_peers.append(peer_id)

            for peer_id in remove_peers:
                del node.peers[peer_id]

    thread = threading.Thread(target=monitor_loop, daemon=True)
    thread.start()
    return thread

```

```

        if remove_peers:
            for peer_id in remove_peers:
                del node.peers[peer_id]
                print(f"\nRemoved inactive peer: {peer_id}")
                node.save_config()

    time.sleep(2)

thread = threading.Thread(target=monitor_loop, daemon=True)
thread.start()
return thread

```

core/config_manager.py

```

import os
import json

class ConfigManager:
    """
    Manages loading and saving peer configurations.
    Configs are saved under `storage/configs/<port>.json` relative to the root directory.
    """
    def __init__(self, port):
        self.port = port
        # Resolve config storage relative to this file's package root
        base_dir = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
        self.storage_dir = os.path.join(base_dir, "storage", "configs")
        self.config_file = os.path.join(self.storage_dir, f"{port}.json")

        # Ensure the storage directory exists
        os.makedirs(self.storage_dir, exist_ok=True)

    def save_config(self, peer_id: str, host: str, registered_nodes: list, peers: dict):
        """Saves current state (registered nodes and discovered peers) to config file."""
        data = {
            "peer_id": peer_id,
            "host": host,
            "port": self.port,
            "registered_nodes": registered_nodes,
            "peers": peers
        }
        with open(self.config_file, "w") as f:
            json.dump(data, f, indent=4)

    def load_config(self) -> dict:
        """Loads and returns config if it exists, otherwise returns None."""
        if os.path.exists(self.config_file):
            with open(self.config_file, "r") as f:
                return json.load(f)
        return None

```

core/__init__.py

```

# Package initialization for core

```

utils/constants.py

```
# Network constants for the P2P multicast system

MULTICAST_GROUP = "224.1.1.1"
MULTICAST_PORT = 5007

BUFFER_SIZE = 4096

DISCOVERY_INTERVAL = 5
PEER_TIMEOUT = 10
```

utils/hashing.py

```
import hashlib

def generate_hash(value: str) -> str:
    """
    Generates a SHA-256 hash for a given string value.
    This hash is used to check the consistency of peer metadata.
    """
    return hashlib.sha256(value.encode()).hexdigest()
```

utils/__init__.py

```
# Package initialization for utils
```

8. Unit Tests

tests/test_hashing.py

```
import sys
import os
import unittest

# Add root folder to sys.path
sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__file__))))

from utils.hashing import generate_hash

class TestHashing(unittest.TestCase):
    def test_hash_consistency(self):
        """Verify that hashing same value multiple times yields identical results."""
        val = "127.0.0.1:5001"
        h1 = generate_hash(val)
        h2 = generate_hash(val)
        self.assertEqual(h1, h2)

    def test_hash_length_and_type(self):
        """Verify that the generated hash is a valid SHA-256 string (64 characters)."""
        val = "test-node"
        h = generate_hash(val)
```

```

        self.assertIsInstance(h, str)
        self.assertEqual(len(h), 64)

    def test_hash_uniqueness(self):
        """Verify that different values produce different hashes."""
        h1 = generate_hash("node1")
        h2 = generate_hash("node2")
        self.assertNotEqual(h1, h2)

if __name__ == "__main__":
    unittest.main()

```

tests/test_discovery.py

```

import sys
import os
import unittest
from unittest.mock import MagicMock, patch

# Add root folder to sys.path
sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__file__))))

from core.discovery import start_discovery_sender, start_discovery_listener

class TestDiscovery(unittest.TestCase):
    @patch('core.discovery.threading.Thread')
    def test_start_discovery_sender(self, mock_thread):
        """Verify that start_discovery_sender starts a background daemon thread."""
        node = MagicMock()
        node.running = True

        thread = start_discovery_sender(node)

        self.assertIsNotNone(thread)
        mock_thread.assert_called_once()
        self.assertTrue(mock_thread.call_args_list[0][1]['daemon'])

    @patch('core.discovery.threading.Thread')
    def test_start_discovery_listener(self, mock_thread):
        """Verify that start_discovery_listener starts a background daemon thread."""
        node = MagicMock()
        node.running = True

        thread = start_discovery_listener(node)

        self.assertIsNotNone(thread)
        mock_thread.assert_called_once()
        self.assertTrue(mock_thread.call_args_list[0][1]['daemon'])

if __name__ == "__main__":
    unittest.main()

```

tests/test_health_monitor.py

```

import sys
import os
import unittest

```

```

import time
from unittest.mock import MagicMock, patch

# Add root folder to sys.path
sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__file__))))

from core.health_monitor import start_health_monitor

class TestHealthMonitor(unittest.TestCase):
    @patch('core.health_monitor.threading.Thread')
    def test_start_health_monitor(self, mock_thread):
        """Verify that start_health_monitor starts a background daemon thread."""
        node = MagicMock()
        node.running = True

        thread = start_health_monitor(node)

        self.assertIsNotNone(thread)
        mock_thread.assert_called_once()
        self.assertTrue(mock_thread.call_args_list[0][1]['daemon'])

    def test_pruning_inactive_peers(self):
        """Verify that health monitor logic correctly identifies and removes timed out peers."""
        node = MagicMock()
        node.running = True
        node.peer_id = "127.0.0.1:5001"

        current_time = time.time()
        node.peers = {
            "127.0.0.1:5001": {
                "host": "127.0.0.1", "port": 5001,
                "nodes": ["node1"], "hash": "hash1",
                "last_seen": current_time          # self - never pruned
            },
            "127.0.0.1:5002": {
                "host": "127.0.0.1", "port": 5002,
                "nodes": ["node2"], "hash": "hash2",
                "last_seen": current_time - 2      # active (2s ago)
            },
            "127.0.0.1:5003": {
                "host": "127.0.0.1", "port": 5003,
                "nodes": ["node3"], "hash": "hash3",
                "last_seen": current_time - 15     # inactive (15s ago > PEER_TIMEOUT)
            }
        }

        # Simulate pruning logic
        remove_peers = []
        for peer_id, info in list(node.peers.items()):
            if peer_id == node.peer_id:
                continue
            if current_time - info["last_seen"] > 10:
                remove_peers.append(peer_id)

        for peer_id in remove_peers:
            del node.peers[peer_id]

        self.assertIn("127.0.0.1:5001", node.peers)
        self.assertIn("127.0.0.1:5002", node.peers)
        self.assertNotIn("127.0.0.1:5003", node.peers)

if __name__ == "__main__":

```

9. README

P2P Multicast System

A decentralized peer-to-peer multicast network system built using Python's standard libraries (socket, threading). It supports dynamic peer discovery via UDP multicast, registered node indexing, and automated heartbeat monitoring for peer health tracking.

Prerequisites

- Python 3.8 or higher
- pytest (optional, for running tests): `pip install -r requirements.txt`

How to Run

Boot a single node:

```
python p2p_multicast_system/main.py
```

Enter a port when prompted (e.g. 5001).

Boot a second node (in a new terminal):

```
python p2p_multicast_system/main.py
```

Enter a different port (e.g. 5002). Both nodes will automatically discover each other.

Commands (inside node shell)

Command	Description
register	Register a new channel/node string for this peer
view nodes	Display all registered nodes across all peers
view peers	Display active peer IPs, ports, and state hashes
exit	Gracefully shut down the node

Running Tests

```
# Using unittest
python -m unittest discover -s p2p_multicast_system/tests

# Using pytest
pytest p2p_multicast_system/
```

10. Requirements

requirements.txt

```
# Testing dependencies
pytest>=7.0.0
```

All runtime functionality uses Python standard library only. No external packages required.

11. Test Results

```
$ python -m unittest discover -s p2p_multicast_system/tests
```

```
.....
```

```
-----  
Ran 7 tests in 0.005s
```

```
OK
```

Test Coverage Summary

Test File	Test Name	What it Verifies	Result
test_hashing.py	test_hash_consistency	Same input -> same SHA-256 output	✓ PASS
test_hashing.py	test_hash_length_and_type	Output is a 64-char string	? PASS
test_hashing.py	test_hash_uniqueness	Different inputs -> different hashes	? PASS
test_discovery.py	test_start_discovery_sender	Sender spawns a daemon thread	? PASS
test_discovery.py	test_start_discovery_listener	Listener spawns a daemon thread	? PASS
test_health_monitor.py	test_start_health_monitor	Monitor spawns a daemon thread	? PASS
test_health_monitor.py	test_pruning_inactive_peers	Peers >10s old are removed; active peers kept	✓ PASS

Total: 7/7 tests passed ?